Supplementary information: PathoFluor-At

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Supplementary Note 1. Assigning genotype and plant labels to segmented seedlings

Each row-scan images one row of seedlings along the scan (line) axis. Genotype lanes are columns along that axis, shared by a plate's three rows. For every plate and timepoint we pooled the line positions of all segmented seedlings and clustered them into five lanes (set A) or three (set B) by exact one-dimensional k-means. Plant numbers follow reading order within a lane, which is how the 2016 hand ROIs are numbered.

Plates were not always placed in the same orientation on the stage. The 2016 hand ROIs show *cml24-2* (lane D) with the highest F_{685}/F_{740} on every set-A plate, while in the segmented images the high-ratio lane changes position between plates. We therefore chose the scan direction per plate and timepoint. For each direction we paired every hand ROI with the seedling that direction assigns to it, and scored the within-plate Spearman correlation of three features (F_{685}/F_{740}, red-peak width, peak intensity). We kept the direction with the higher mean (Table S1). Plates without hand ROIs (set-A Pathogen at 48 h) receive no genotype label and enter only treatment-level analyses. Set-B hand ROIs are per leaf, so set-B lanes follow the order stated in the vendor report (Col-0, *fls2-17*, *fls2-101*; FACTS R9), pending confirmation.

Across set A, 68 of 72 plate-timepoints had hand ROIs to choose a direction from; 16 choices had a margin below 0.2 and are flagged in `results/label_direction_scores.csv`. For set B, the vendor's lane order agreed better with the plate-1 leaf ROIs than the reverse in 21 of 24 hour $\times$ treatment $\times$ laser comparisons.

Supplementary Note 2. The 2016 hand-drawn ROIs

The 2016 annotator drew ROIs on seedlings and exported their mean spectra. Copies were sorted by timepoint, by genotype, and into curated "USE" subsets (1,358 set-A spectra), plus 16 Windows "- Copy" duplicates. Identical copies were merged by content hash. 12 spectra share a design key with a different spectrum (re-drawn ROIs); both are kept and flagged. ROIs carry 25 distinct free-text tags ("bright", "norm", "dark", "side", "dying" and others) whose meaning is pending (FACTS P6). They are retained in `results/roi_features.csv.gz`. No ROIs exist for set-A Pathogen plates at 48 h.

Supplementary Note 3. The 2016 vendor analyses

Two Middleton Spectral Vision reports (H. Jones, 2016 and 2017; unpublished) analysed subsets of these data by multivariate curve resolution. For the July set (one infected and one control plate, two plants per genotype) the report describes two chlorophyll-like components and a broad autofluorescence component present mainly in plants infected for at least 24 h, with infected and control plants separated by component medians. For the November set (one plate per treatment) it describes a "chlorophyll shift" component that rises in *Pseudomonas* plants from 6 h and in flg22-treated Col-0 but not *fls2*, tested with notched box plots of individual plants. Neither analysis treated the plate as the experimental unit or used a pre-treatment baseline. We treat

their conclusions as hypotheses, tested in the main text. The reports are cited, not redistributed (FACTS R19).

Supplementary Note 4. Pilot experiment

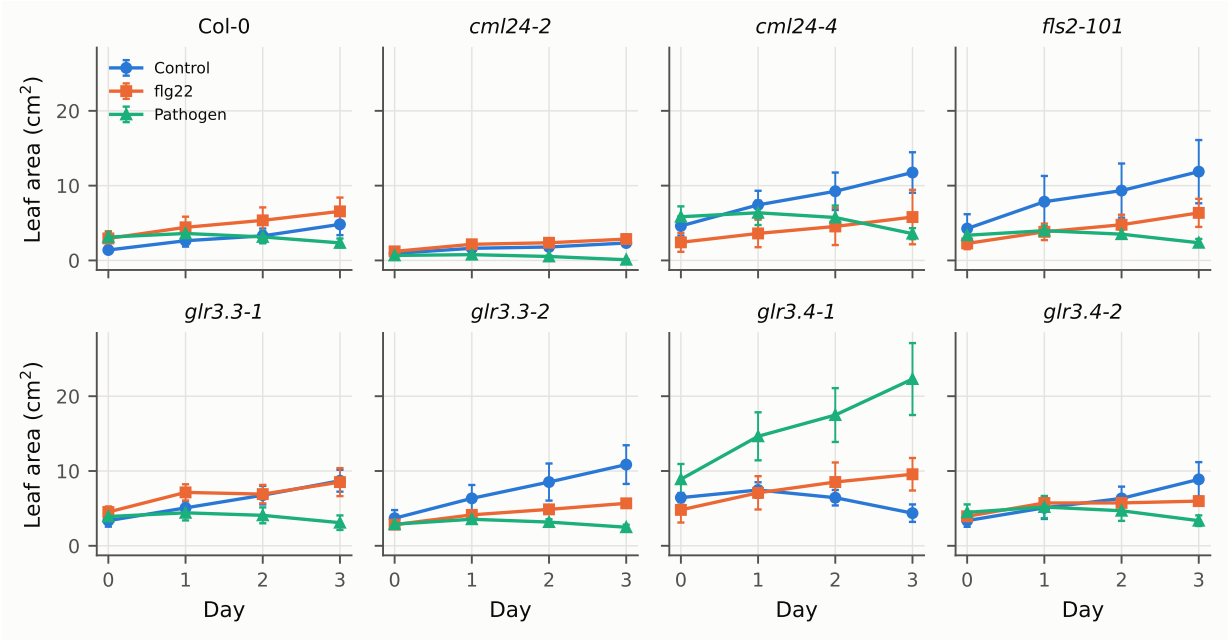


Figure S1: Pilot (February–March 2016) leaf area, days 0–3, in units of a red reference square of 4 cm^2 (FACTS R18), converted to cm^2. 98.3% of the tabulated values match the reference-ratio column of the source sheet (FACTS R10). Data: `tables/figS1_pilot_leaf_area.csv`; source sheet in `data/related_drive_2016/`.

Supplementary Note 5. Towards practical monitoring

Two analyses ask what the result would mean for a working scanner. Both were pre-specified: `scripts/11_field_analyses.py` was committed before its first run. They use the Col-0 baseline-relative hand-ROI spectra, the L2 logistic model and the leave-one-plate-out validation of the main analysis.

Monitoring replay. At each hour, every Pathogen and Control plate was scored by a model fitted without it, and a plate was flagged when its mean held-out probability of infection exceeded 0.5 (Fig. S2). 3 of the 4 Pathogen plates were flagged from 1 h and the last at 24 h (13 of 16 Pathogen plate-hours). 0 of 12 Control plate-hours were flagged. The 3 flg22 plates, never used in training, were flagged at 9 of 9 plate-hours at 1–3 h and at 0 of 3 at 24 h, when 4 of 4 Pathogen plates were flagged. The early signature is shared by the elicitor treatment and the bacterium; only the bacterial one persisted. With three or four plates per arm this is a demonstration, not a validated detector.

Filter bands. Each spectrum was reduced to the mean signal in 10 nm boxcar filters at 685 and 740 nm, at 670, 685 and 700 nm, or at all four, expressed as log ratios to the 685 nm band, as a filter camera would record them. For Pathogen vs Control, the best plate AUC of any filter set at any hour was 0.75 (smallest permutation $p = 0.057$), against 0.75–1.00 for the full 2 nm spectrum; the F685/F740 ratio alone reached at most 0.67 (Fig. S3). The Pathogen signal is a broadening of

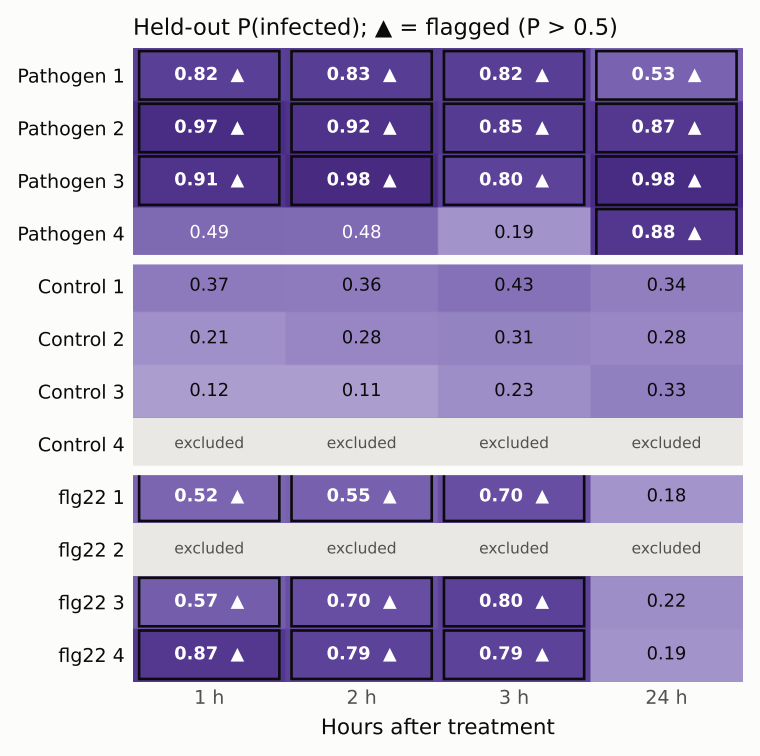


Figure S2: Monitoring replay: held-out probability of infection per plate and hour. Pathogen and Control plates are scored leave-one-plate-out; flg22 plates by the model fitted on all Pathogen and Control plates. A triangle and box mark a flag ($P > 0.5$). Control-4 and flg22-2 are excluded because their hour-0 scans were partly upright rescans; Pathogen 48 h has no hand ROIs. Data: `tables/alarm_replay.csv`.

the red peak by about 1.40–1.76 nm (main text), which 10 nm filters average away. A practical sensor would need spectrometer-class resolution across 650–760 nm rather than a few filters.

The cube in three dimensions. Fig. S4 shows one row-scan as a hyperspectral cube. An interactive version, with every timepoint for Control and Pathogen plate 1, is on the project page.

Supplementary Table: strongest baseline-adjusted contrasts

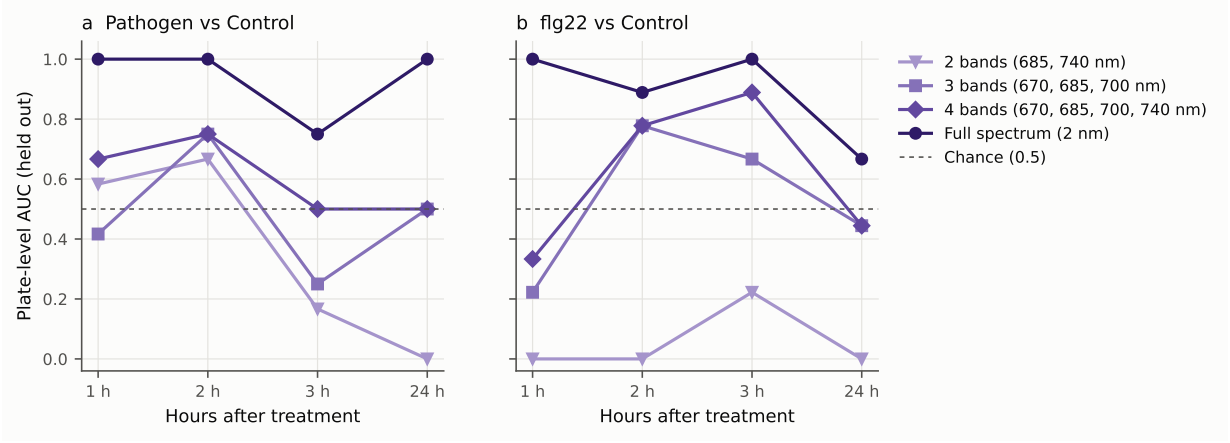


Figure S3: Plate-level held-out AUC for pre-specified filter-band sets and for the full spectrum (Col-0, baseline-relative). Dashed line: chance. Values below 0.5 are held-out reversals typical of few plates, not real reversed signals. Data: `tables/band_sets_auc.csv`, which also gives the exact permutation p for every point.

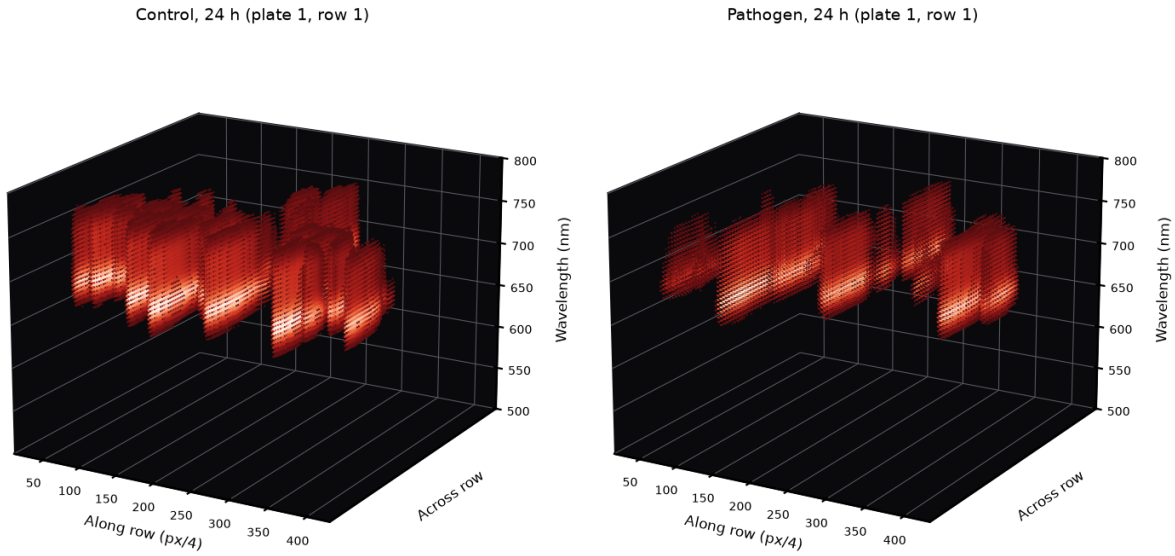


Figure S4: One row-scan (plate 1, row 1, 24 h) as a cube of position along the row, across the row and emission wavelength. Each point is a 4×4 block of plant pixels at one 5 nm band, coloured by exposure-corrected emission; dim points are omitted. Data: `docs/data/cube/` (`scripts/12_site_media.py`).

Table S1: Scan direction per set-A plate and timepoint, chosen by agreement between the 2016 hand ROIs and the seedlings each direction assigns to them (mean within-plate Spearman correlation of three features). Source: results/label_direction_scores.csv.

Hour	Treatment	Plate	ROIs	Seedlings	Score fwd	Score rev	Chosen
0	Control	1	29	28	0.78	-0.36	fwd
0	Control	2	30	27	0.50	0.48	fwd
0	Control	3	30	23	0.28	0.17	fwd
0	Control	4	28	23	-0.06	-0.61	fwd
0	Pathogen	1	27	24	0.44	0.05	fwd
0	Pathogen	2	26	29	-0.32	0.58	rev
0	Pathogen	3	27	26	-0.10	0.35	rev
0	Pathogen	4	30	28	0.27	0.06	fwd
0	flg22	1	29	28	0.84	0.37	fwd
0	flg22	2	30	23	0.07	0.28	rev
0	flg22	3	30	28	0.22	0.24	rev
0	flg22	4	29	22	-0.10	0.04	rev
1	Control	1	29	25	0.13	0.46	rev
1	Control	2	30	24	0.47	-0.08	fwd
1	Control	3	30	21	-0.29	0.19	rev
1	Control	4	28	23	0.12	-0.47	fwd
1	Pathogen	1	28	19	0.04	0.32	rev
1	Pathogen	2	29	26	0.83	0.09	fwd
1	Pathogen	3	29	23	-0.02	0.01	rev
1	Pathogen	4	26	20	-0.33	0.34	rev
1	flg22	1	29	28	0.21	0.38	rev
1	flg22	2	29	23	0.44	-0.03	fwd
1	flg22	3	29	28	0.01	-0.06	fwd
1	flg22	4	30	21	0.36	-0.07	fwd
2	Control	1	30	24	0.15	0.29	rev
2	Control	2	30	24	0.63	-0.06	fwd
2	Control	3	30	21	-0.21	0.28	rev
2	Control	4	29	22	-0.16	-0.43	fwd
2	Pathogen	1	30	19	-0.25	0.27	rev
2	Pathogen	2	29	26	0.82	-0.14	fwd
2	Pathogen	3	30	23	-0.16	-0.20	fwd
2	Pathogen	4	26	21	-0.06	0.42	rev
2	flg22	1	26	26	0.05	0.73	rev
2	flg22	2	30	22	0.18	-0.23	fwd
2	flg22	3	30	27	0.07	0.08	rev
2	flg22	4	30	21	0.23	-0.07	fwd
3	Control	1	30	24	0.19	0.38	rev
3	Control	2	30	25	0.67	0.01	fwd
3	Control	3	29	22	-0.21	0.12	rev
3	Control	4	28	22	0.19	-0.18	fwd
3	Pathogen	1	30	19	-0.32	0.28	rev
3	Pathogen	2	30	25	0.37	-0.13	fwd
3	Pathogen	3	30	23	-0.10	0.23	rev
3	Pathogen	4	28	20	0.27	0.07	fwd
3	flg22	1	29	26	-0.04	0.36	rev
3	flg22	2	30	22	0.32	-0.06	fwd
3	flg22	3	28	26	0.05	0.12	rev
3	flg22	4	30	21	0.26	-0.12	fwd
24	Control	1	30	19	0.03	0.26	rev
24	Control	2	30	22	0.45	-0.00	fwd
24	Control	3	30	19	-0.24	0.01	rev
24	Control	4	29	20	0.13	-0.18	fwd
24	Pathogen	1	30	20	-0.38	0.19	rev
24	Pathogen	2	29	24	0.60	-0.18	fwd
24	Pathogen	3	29	24	-0.13	0.24	rev
24	Pathogen	4	25	18	-0.00	-0.02	fwd
24	flg22	1	30	23	0.01	0.47	rev
24	flg22	2	30	15	0.39	-0.02	fwd
24	flg22	3	30	21	-0.10	-0.05	rev

Table S2: Baseline-adjusted contrasts with $p < 0.01$ (set A; all 270 tests in tables/stats_contrasts_baseline_adjusted.csv).

Hour	Feature	Contrast	Effect (SD)	Effect (units)	p	q
24	shortwave_fraction	Pathogen vs Control (Col-0)	2.46	0.05944	1.6e-07	0.000
2	red_fwhm_nm	Pathogen vs Control (Col-0)	2.39	1.762	5.0e-06	0.001
3	shortwave_fraction	Pathogen vs Control (Col-0)	2.03	0.0006432	7.6e-06	0.001
24	red_fwhm_nm	Pathogen vs Control (Col-0)	2.23	3.348	7.4e-05	0.005
3	red_fwhm_nm	Pathogen vs Control (Col-0)	2.07	1.017	9.8e-05	0.005
3	red_fwhm_nm	flg22 vs Control (Col-0)	2.18	1.074	1.2e-04	0.005
1	red_fwhm_nm	Pathogen vs Control (Col-0)	2.07	1.404	5.1e-04	0.018
2	shortwave_fraction	Pathogen vs Control (Col-0)	1.77	0.0004837	5.3e-04	0.018
1	farred_fraction	Pathogen vs Control (Col-0)	1.94	0.02878	1.2e-03	0.033
2	farred_fraction	Pathogen vs Control (Col-0)	1.86	0.02795	1.2e-03	0.033
24	shortwave_fraction	Pathogen x cml24-4	-1.74	-0.042	1.4e-03	0.036
48	red_fwhm_nm	flg22 vs Control (Col-0)	1.49	0.9618	1.7e-03	0.037
2	centroid_nm	Pathogen vs Control (Col-0)	1.82	1.667	2.3e-03	0.048
1	centroid_nm	Pathogen vs Control (Col-0)	1.88	1.706	2.7e-03	0.052
2	f685_f740	Pathogen vs Control (Col-0)	-1.74	-0.8395	4.0e-03	0.073
2	red_fwhm_nm	flg22 vs Control (Col-0)	1.60	1.177	4.4e-03	0.074
1	f685_f740	Pathogen vs Control (Col-0)	-1.75	-0.8463	7.2e-03	0.114
24	shortwave_fraction	Pathogen x cml24-2	-1.41	-0.03421	9.5e-03	0.132
48	farred_fraction	flg22 vs Control (Col-0)	1.29	0.01529	9.5e-03	0.132
48	f685_f740	flg22 vs Control (Col-0)	-1.26	-0.5149	9.8e-03	0.132